

Bivalves



Step 1: Data exploration

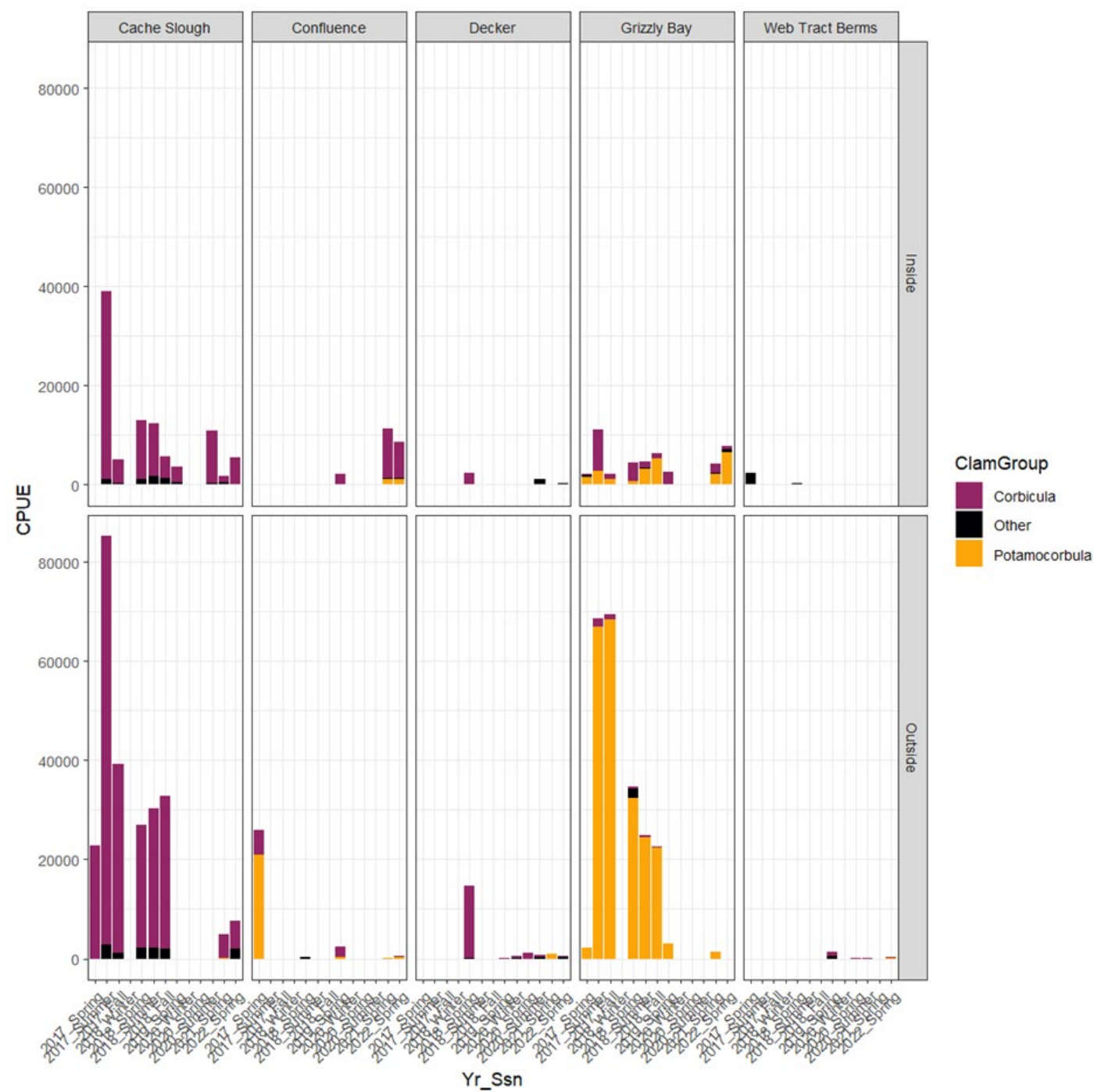
Samples per region and type, all spp



CPUE per region and type, by spp

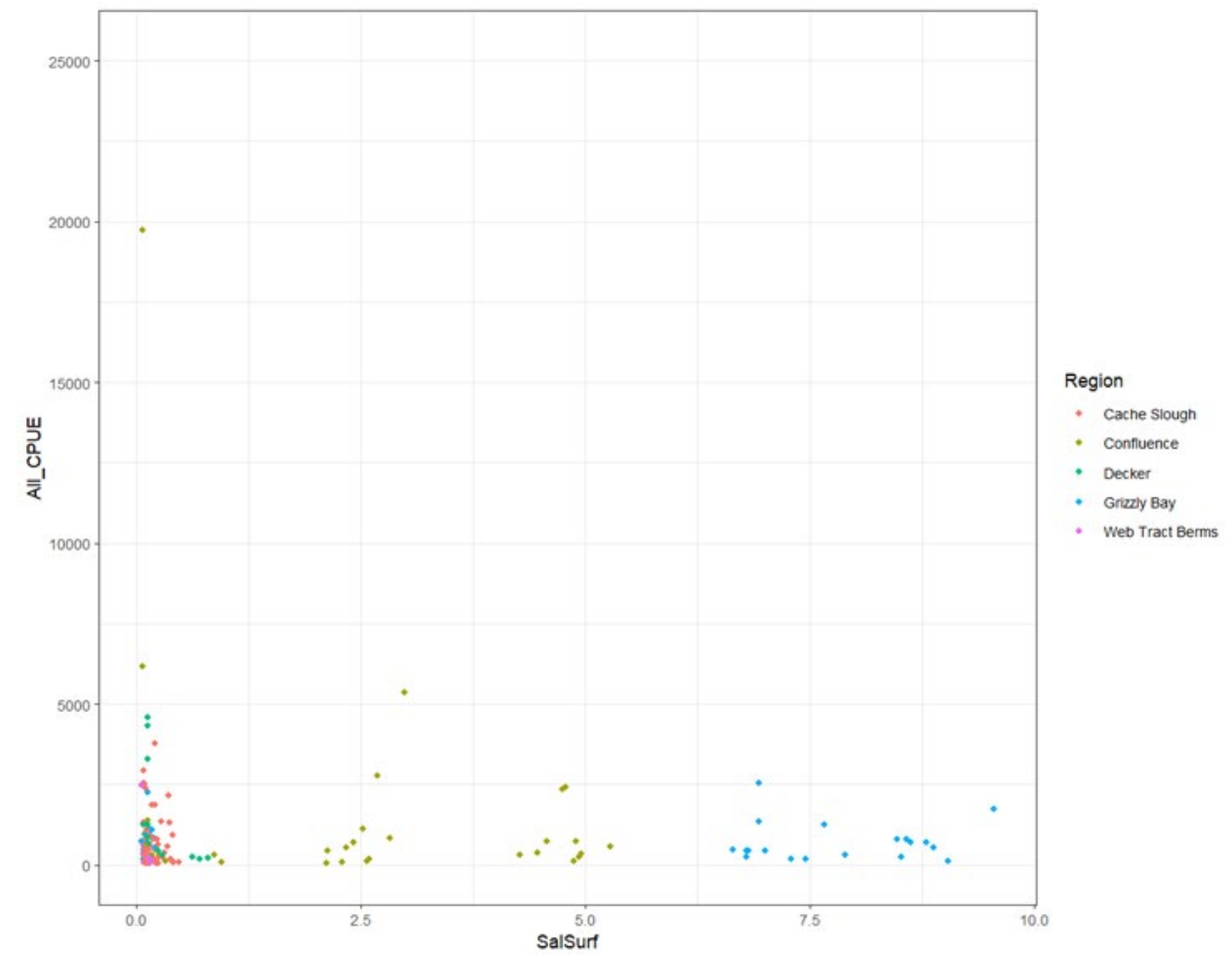
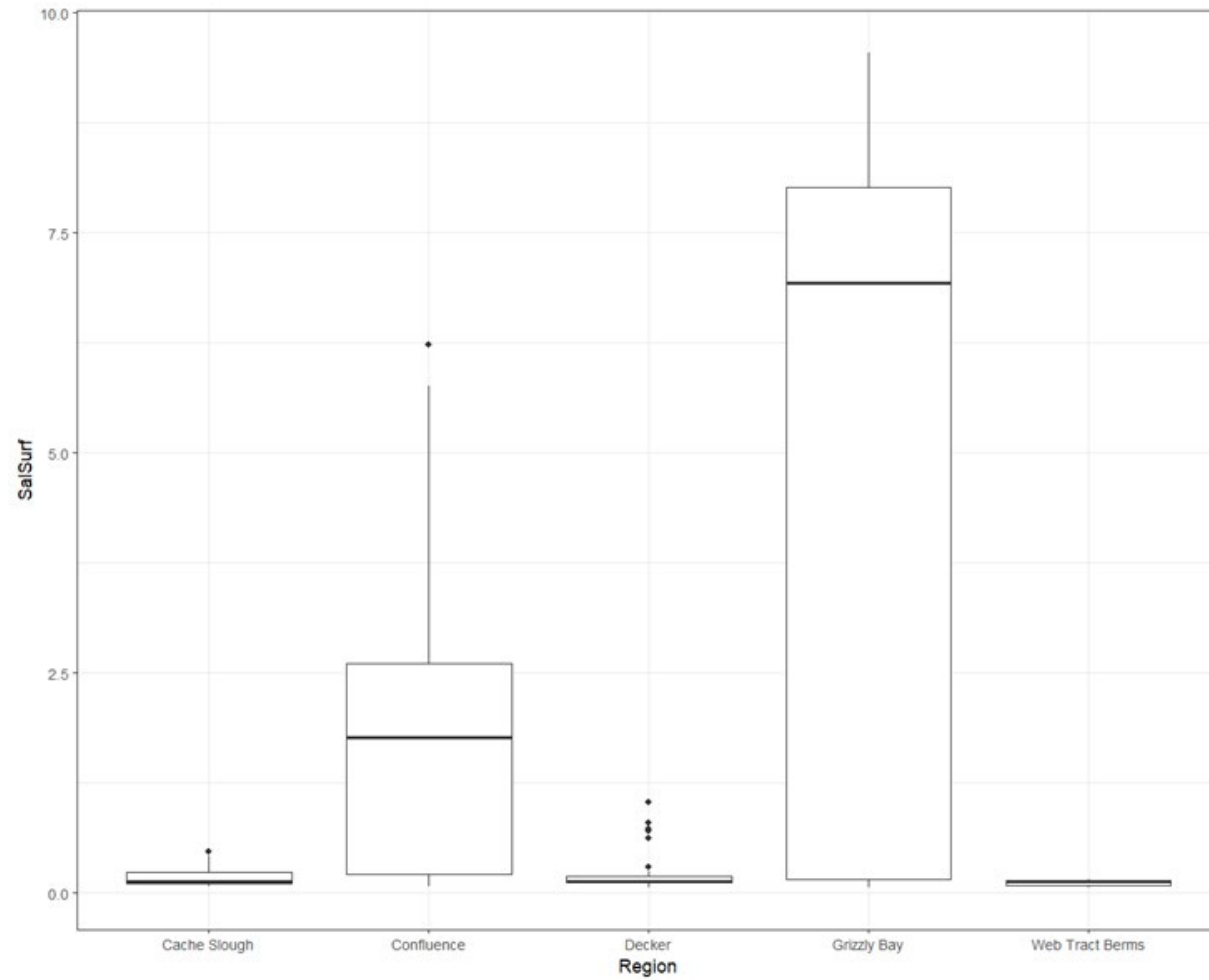
As expected, Corbicula and Potamocorbula dominant in different regions

- Relatively few other taxa detected, especially outside of Cache



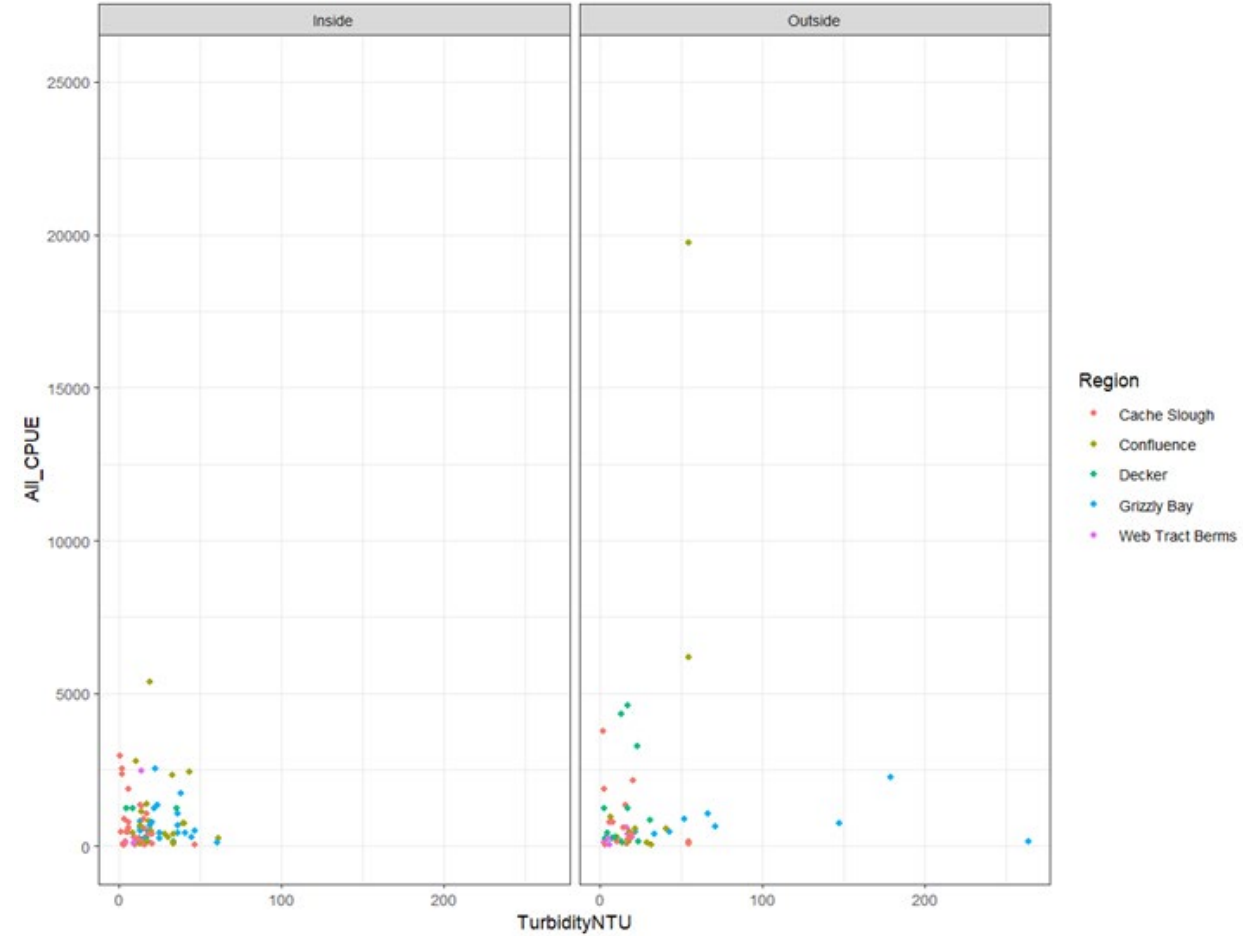
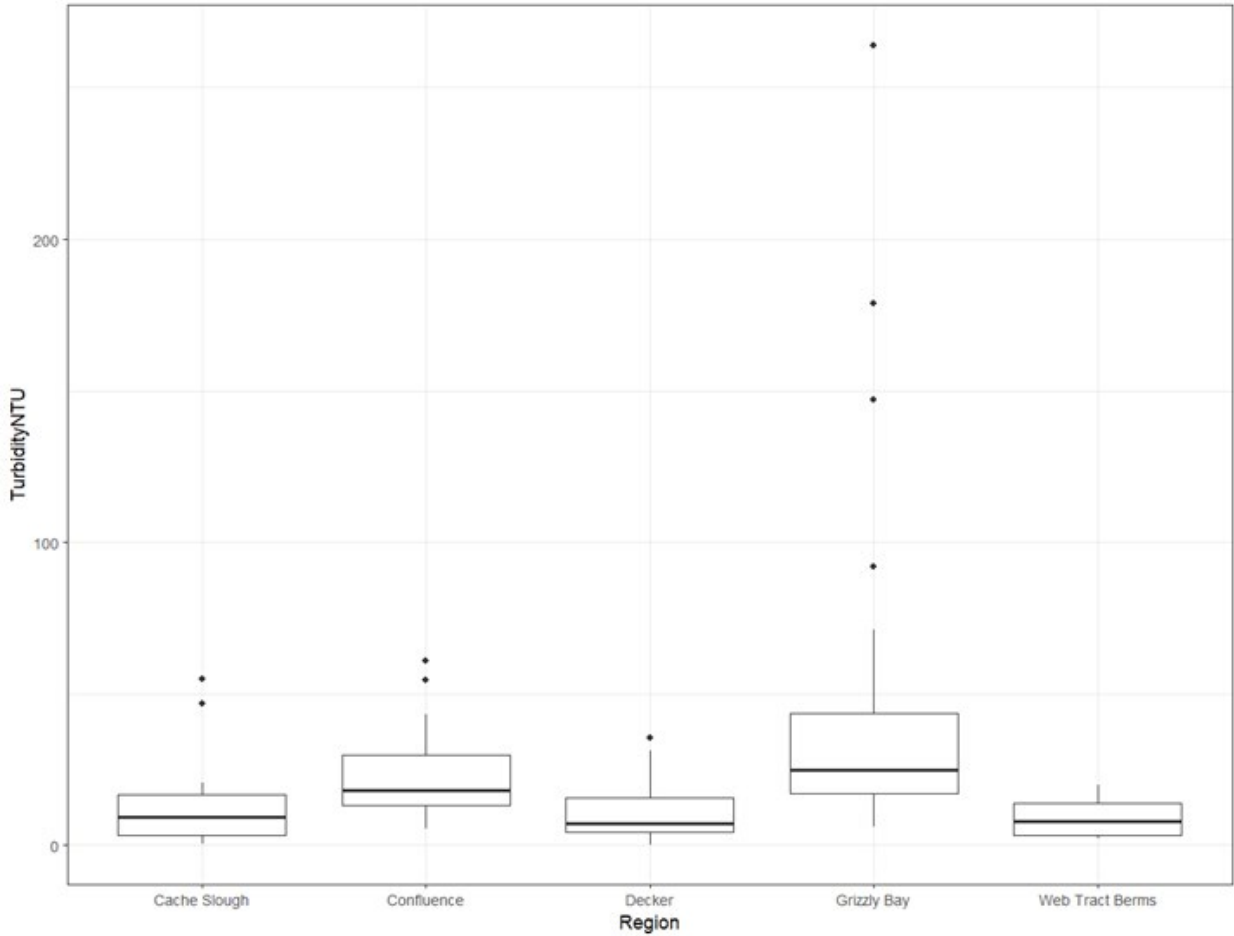
Salinity: region, CPUE (all spp)

No obvious relationship between CPUE of all taxa and salinity, inside and outside combined



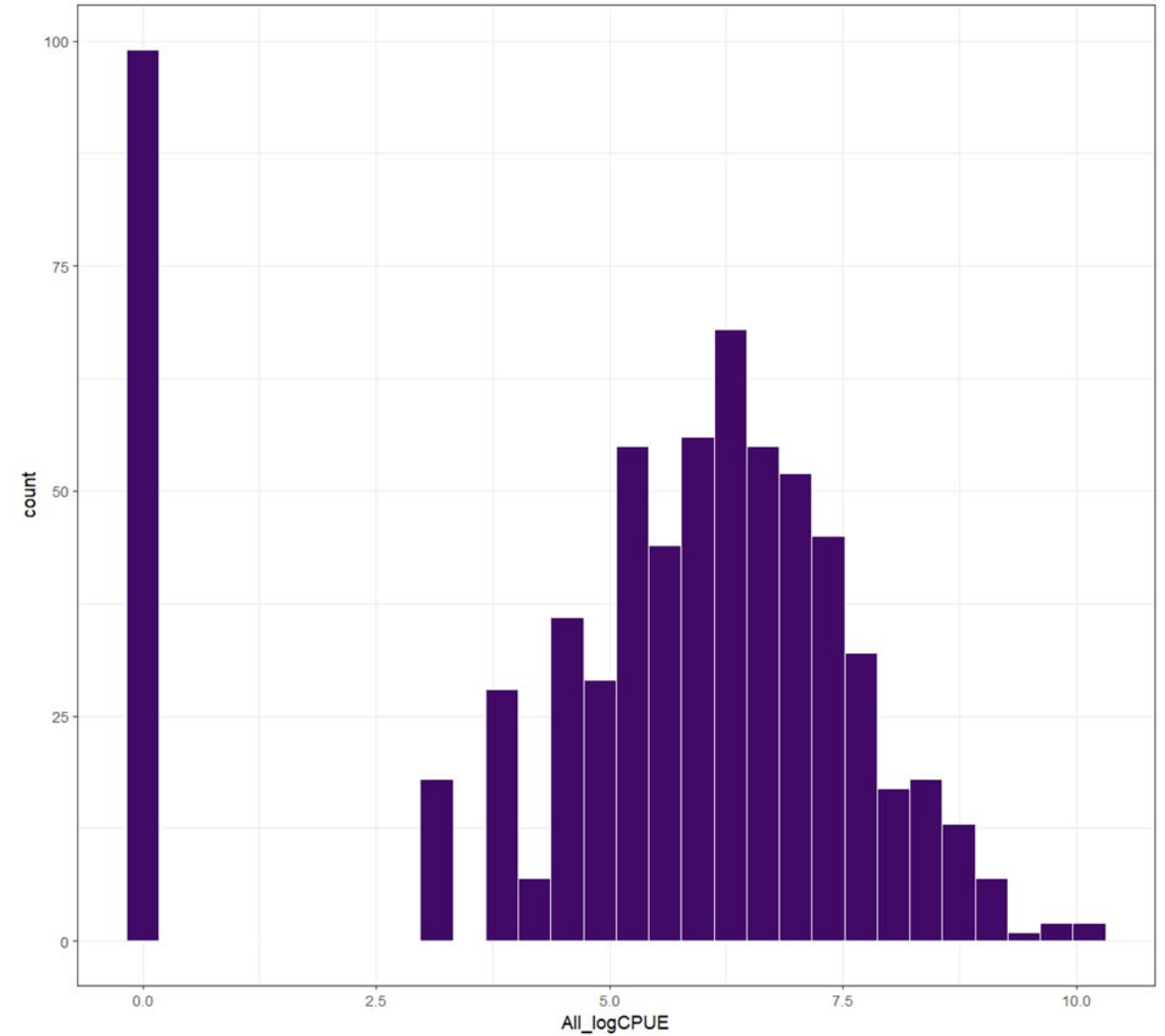
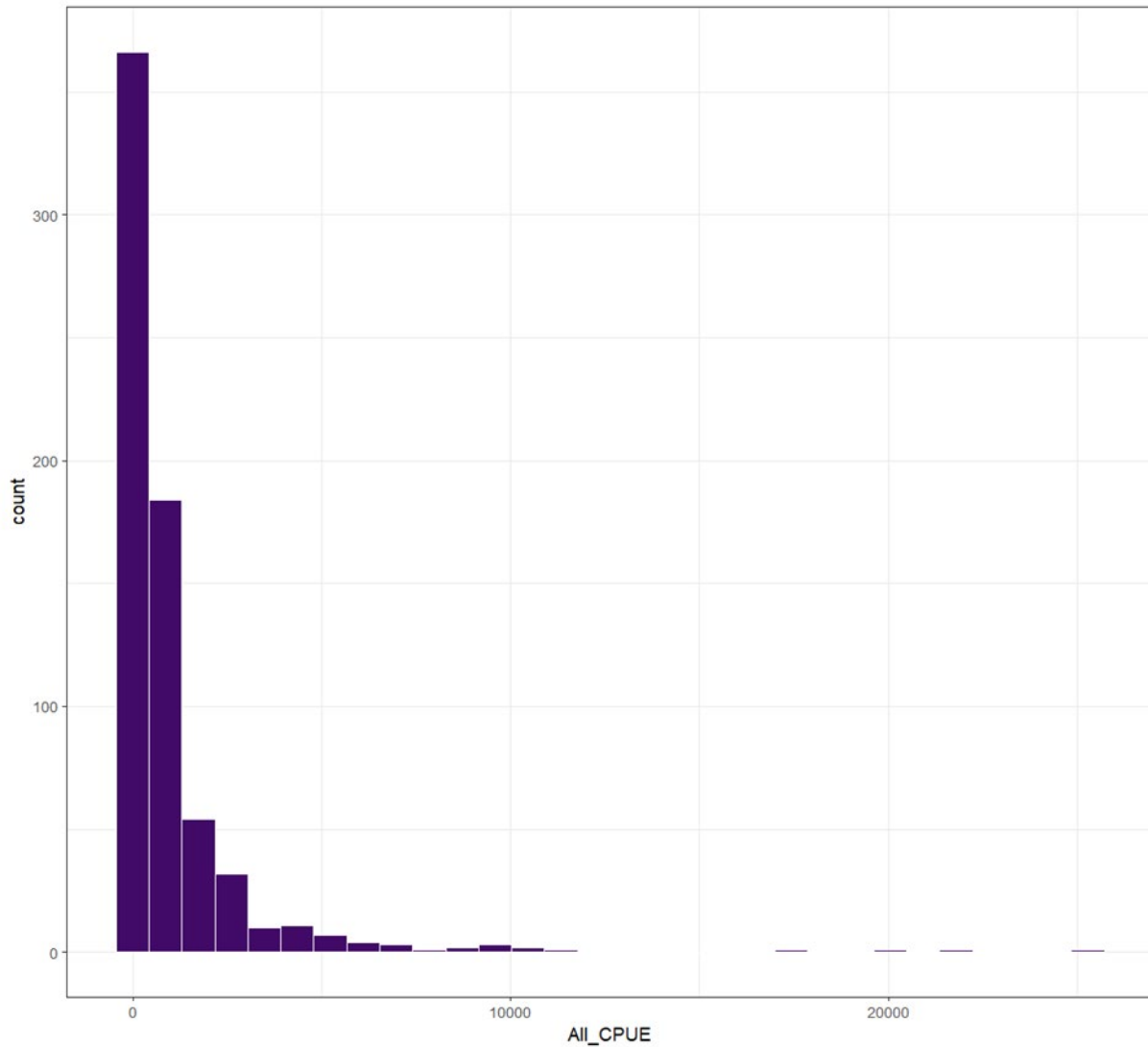
Turbidity: region, CPUE (all spp)

No obvious relationship between CPUE of all taxa and turbidity, inside and outside combined



Distribution: CPUE (all spp)

Hurdle model with gamma or lognormal distribution of positive CPUEs seems appropriate



Step 2: Initial model exploration

CPUE (all spp): gamma vs. lognormal models

Spoiler! lognormal hurdle models outperformed gamma hurdle models

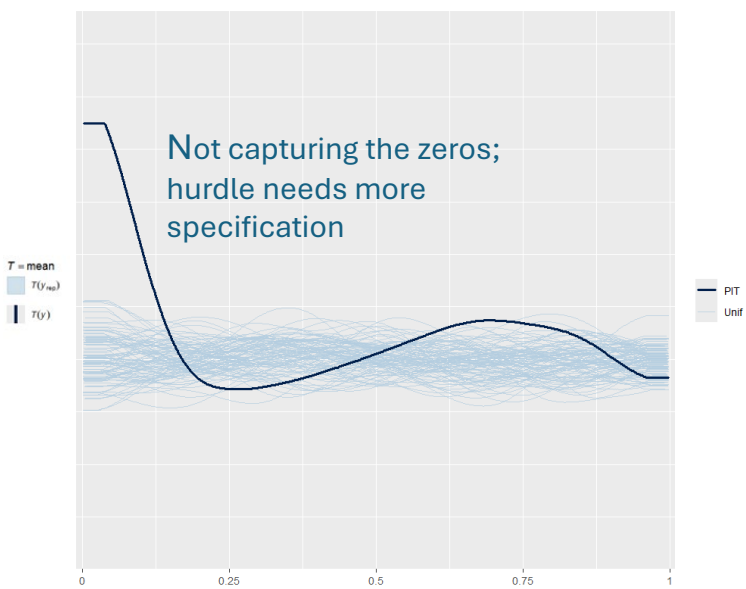
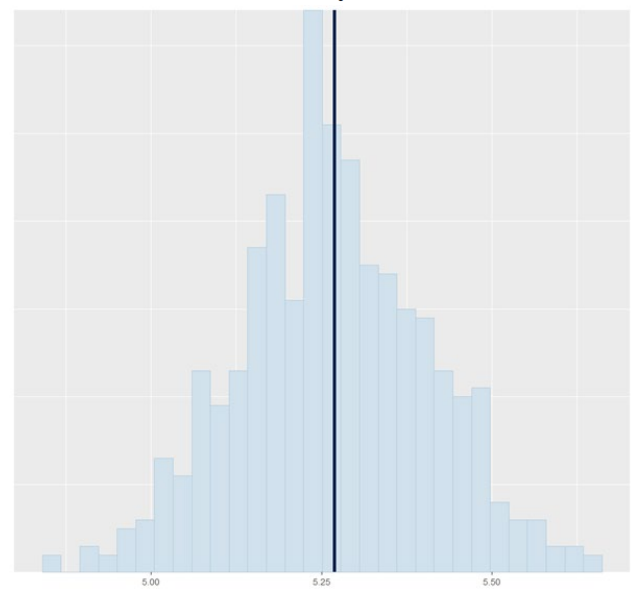
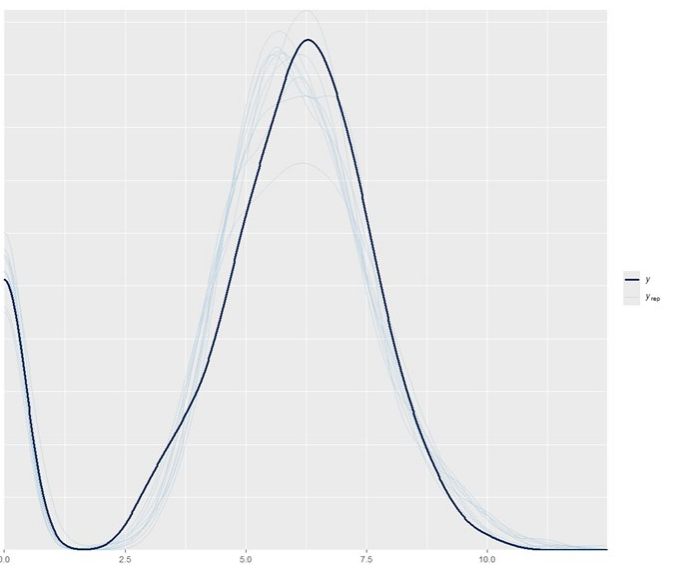
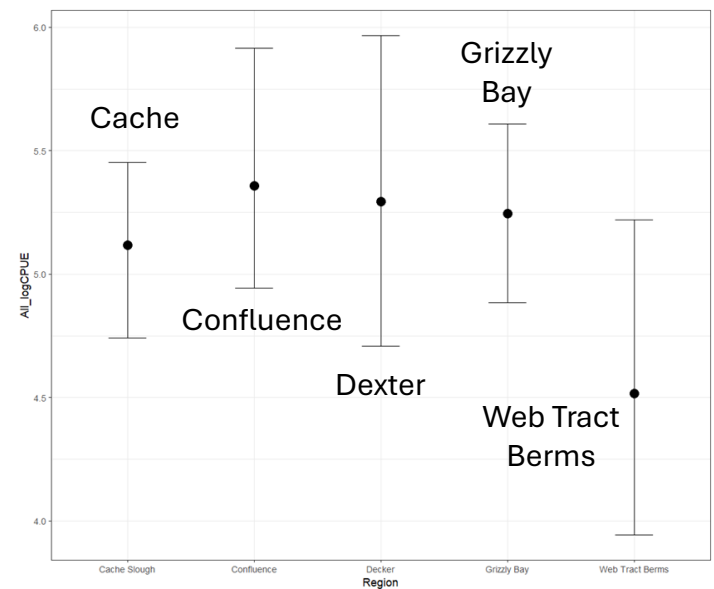
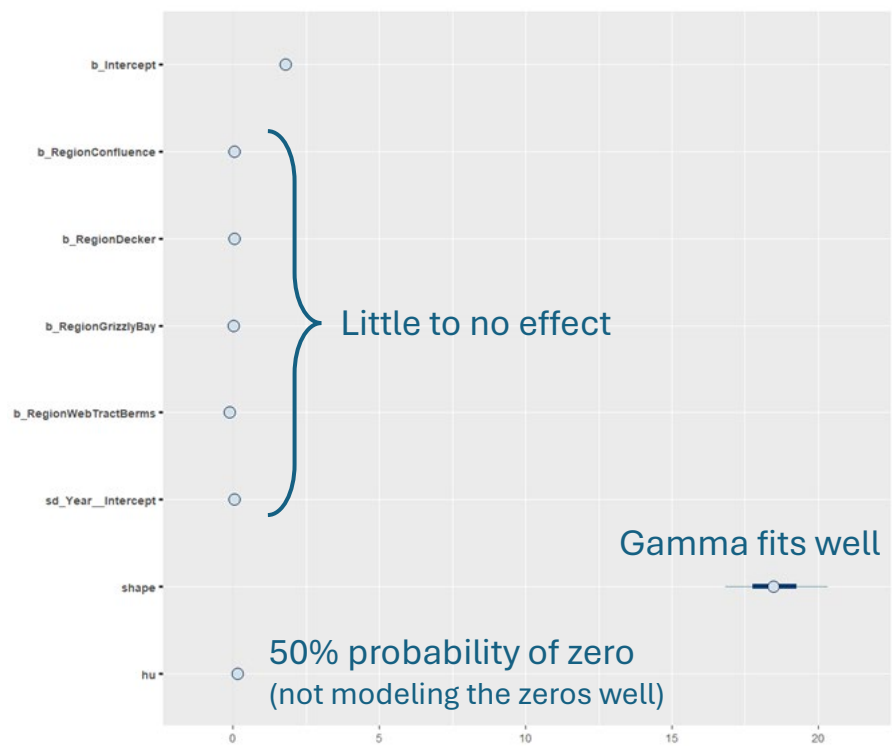
...and are comparable to each other

		ELPD_diff	se_diff	
Lognormal hurdle	All_CPUE ~ Region + (1 Year) ; hu~1	0.0	0.0	
	All_CPUE ~ Type + (1 Year) ; hu~1	-1.1	2.4	-1.1-(-46.7) >? 1.96*10.5 → 45.6 > 20.58
Gamma hurdle	All_CPUE ~ Region + (1 Year); hu~1	-46.7	10.5	
	All_CPUE ~ Type + (1 Year) ; hu~1	-48.1	11.4	

CPUE (all spp): Simple gamma models

Model 01: All_CPUE ~ Region + (1 | Year)

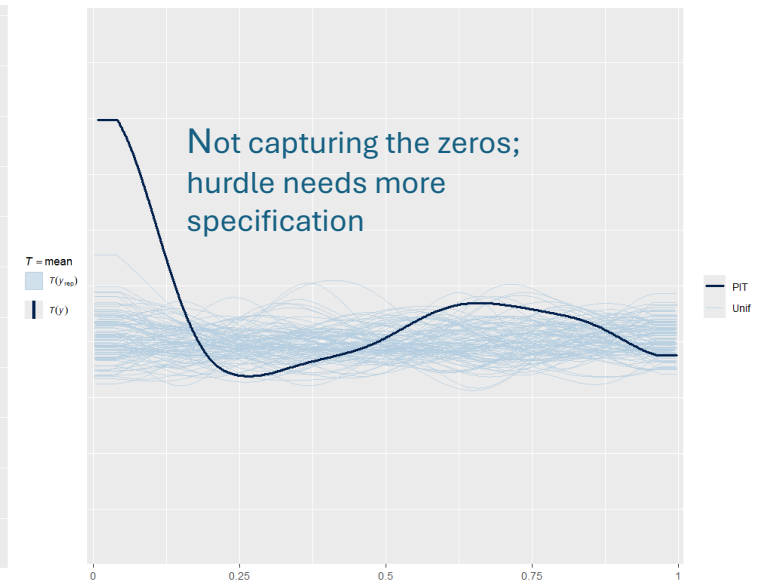
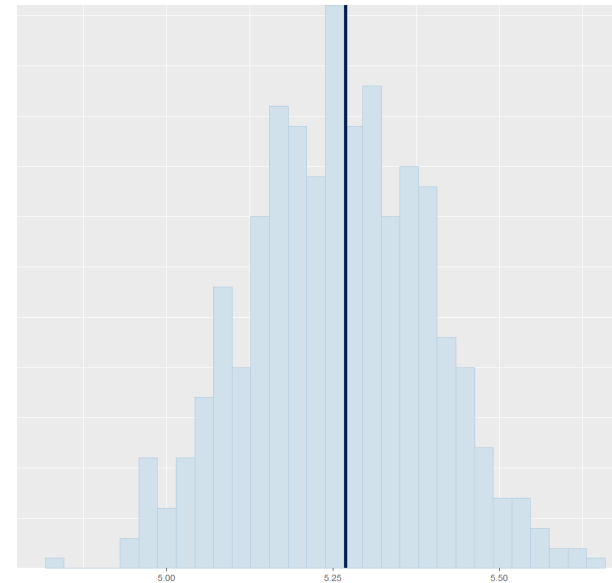
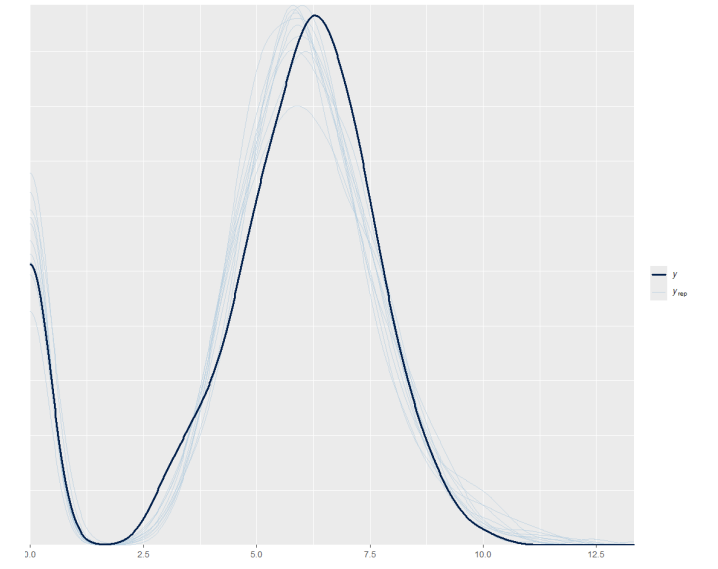
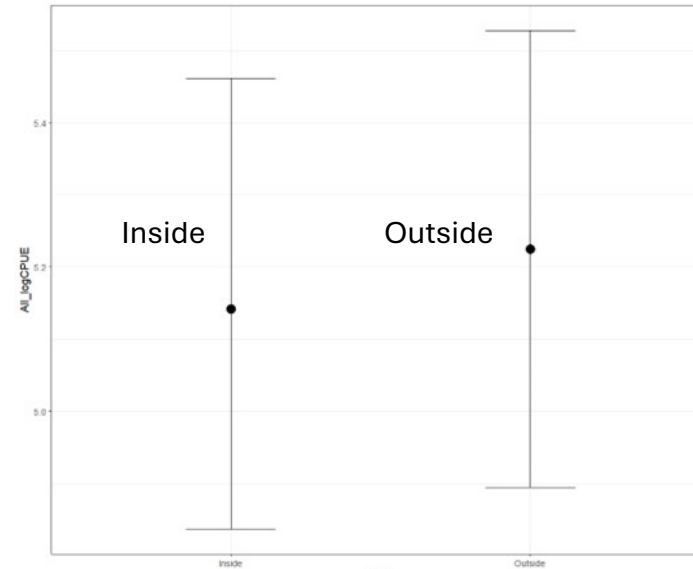
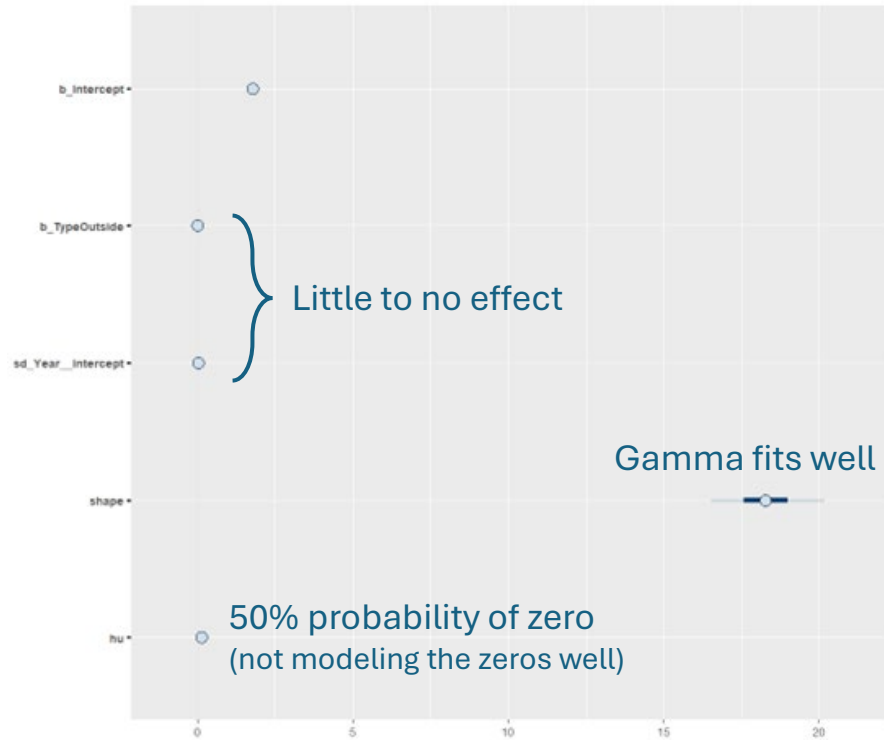
hu ~ 1



CPUE (all spp): Simple gamma models

Model 02: All_CPUE ~ Type + (1 | Year)

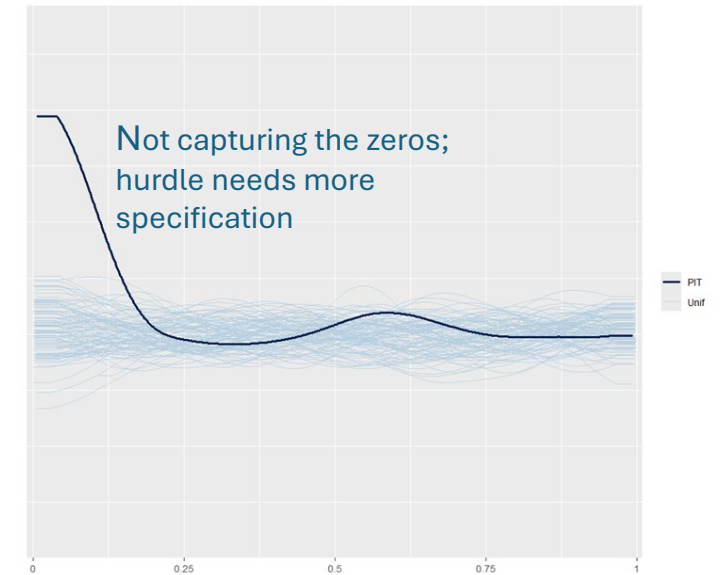
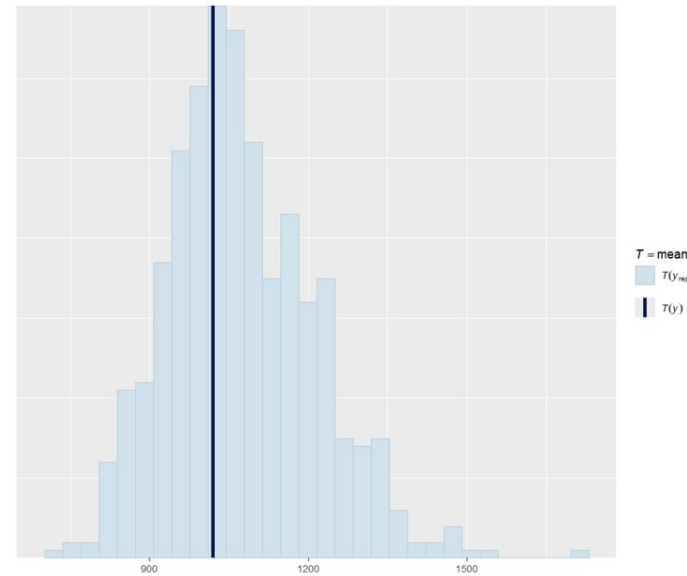
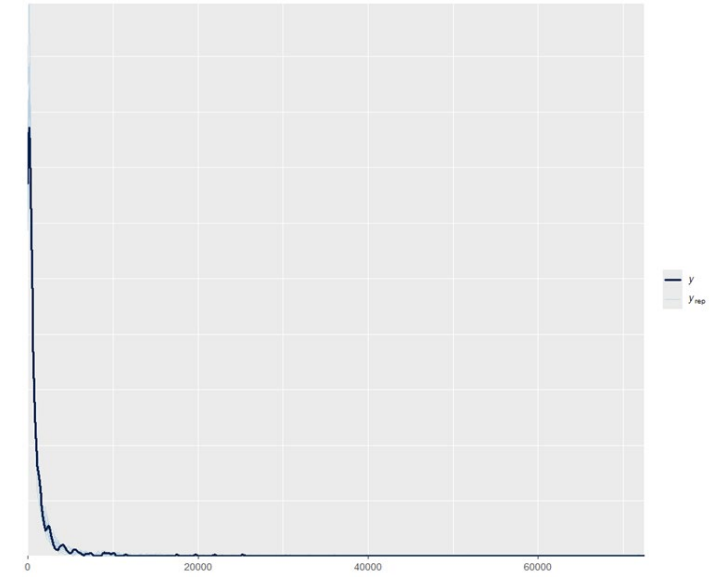
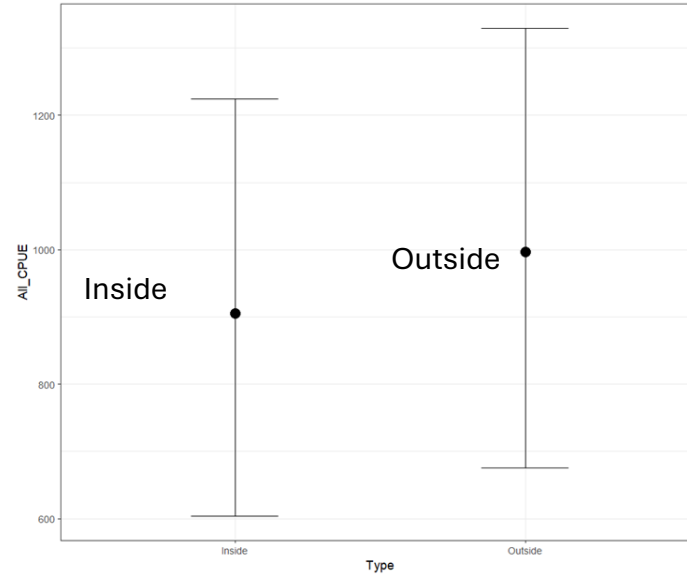
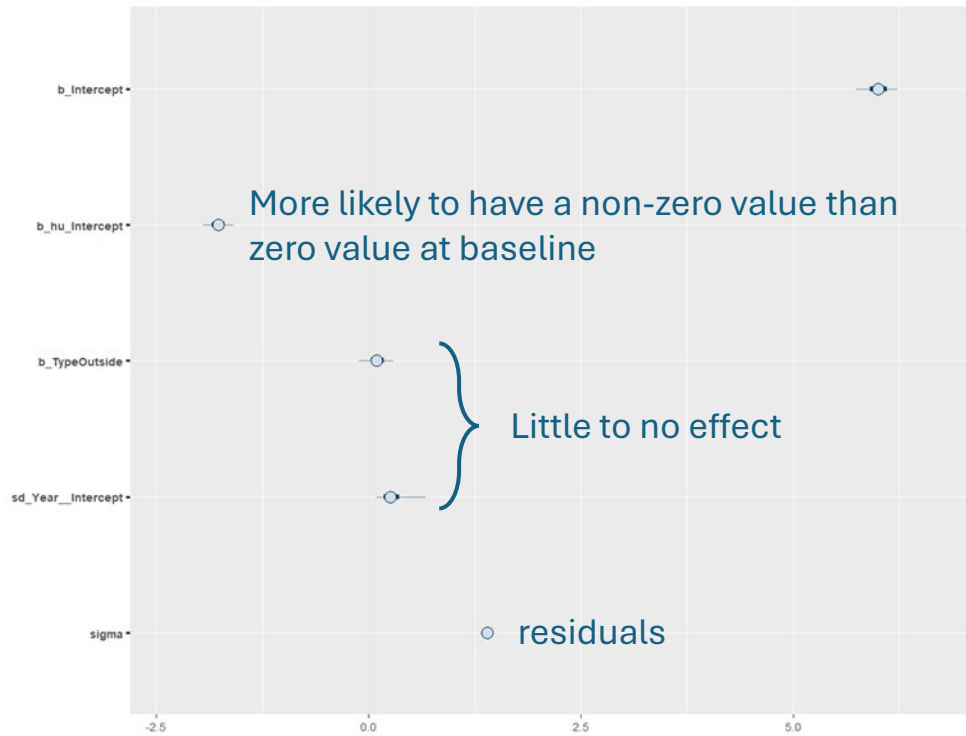
hu ~ 1



CPUE (all spp): Simple lognormal models

Model 11: All_CPUE ~ Type + (1 | Year)

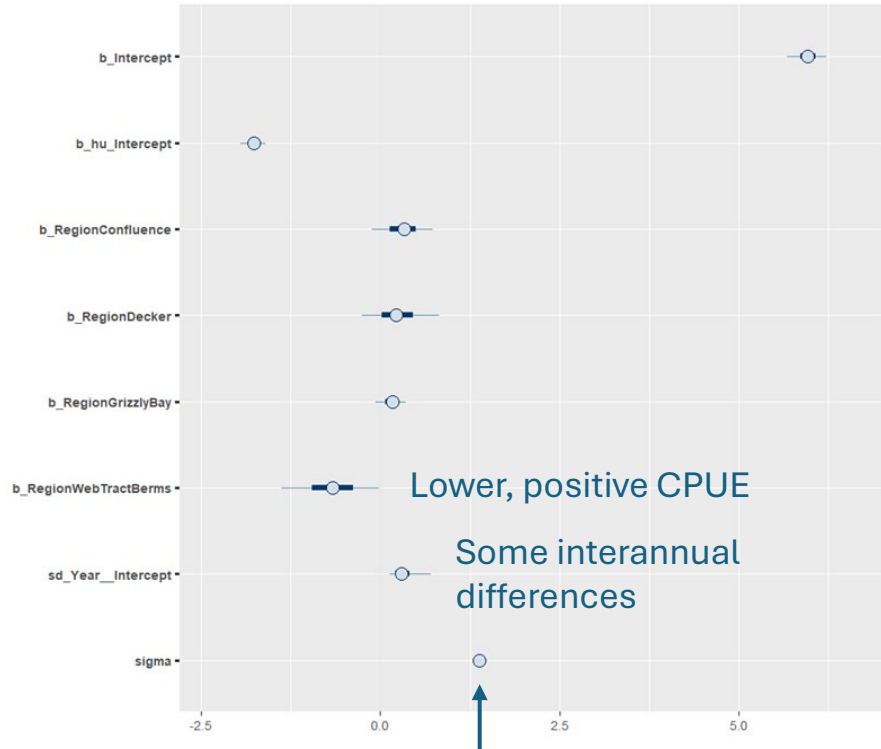
hu ~ 1



CPUE (all spp): Simple lognormal models

Model 12: All_CPUE ~ Region + (1 | Year)

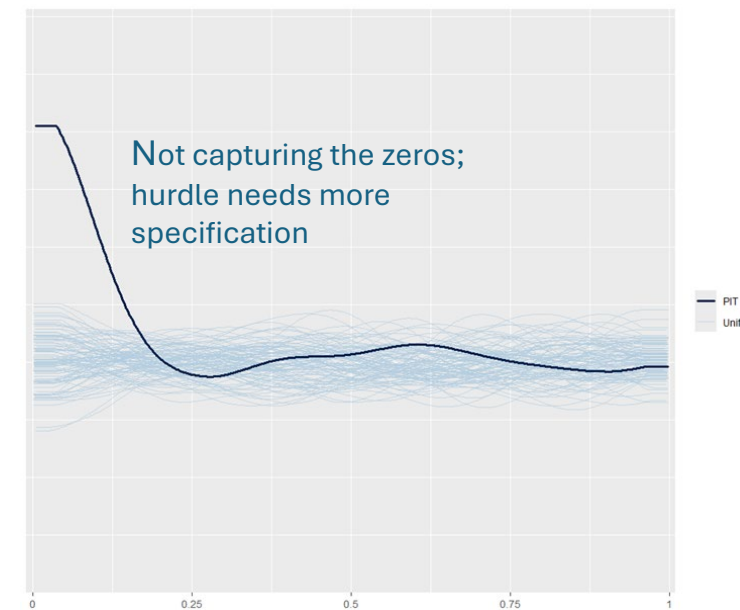
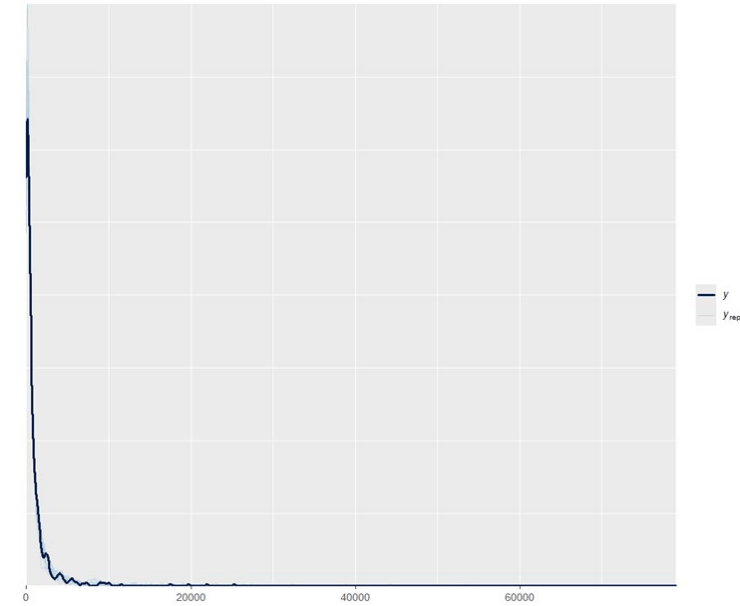
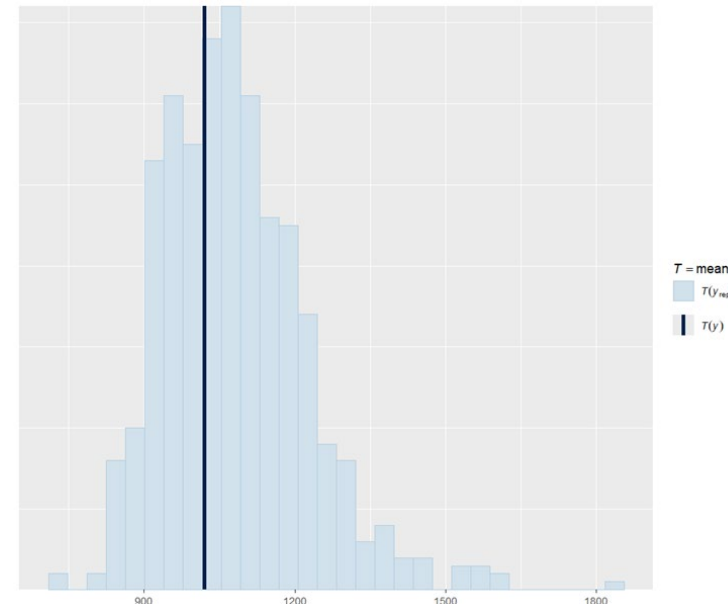
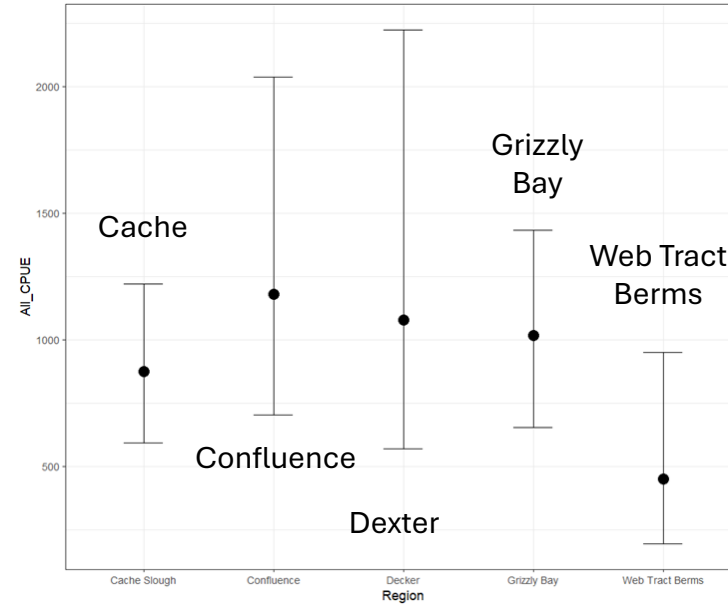
hu ~ 1



Lower, positive CPUE

Some interannual differences

Noticeable variability in positive CPUE after accounting for region and year

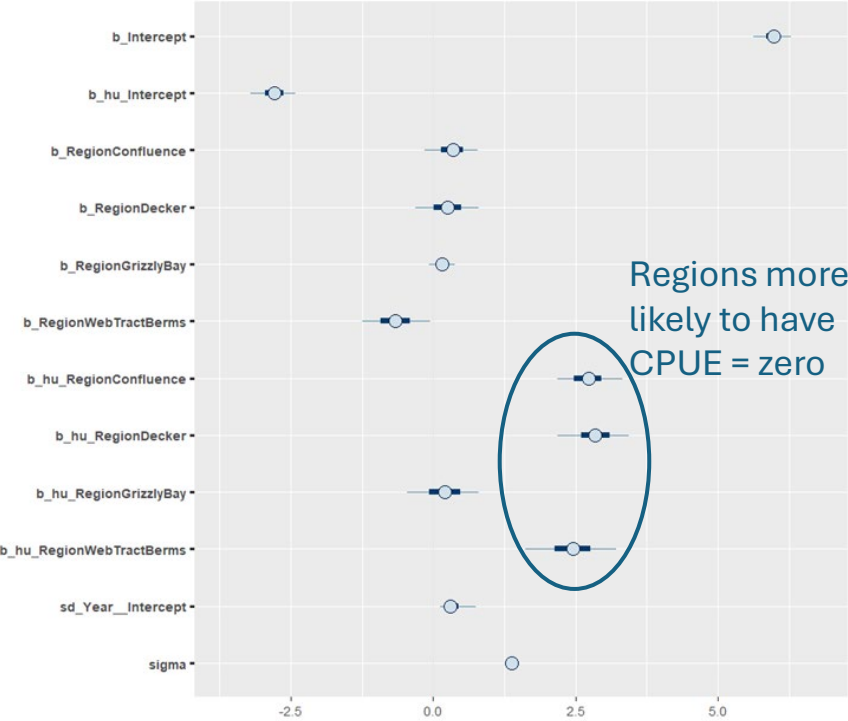


Step 3: Model alternatives

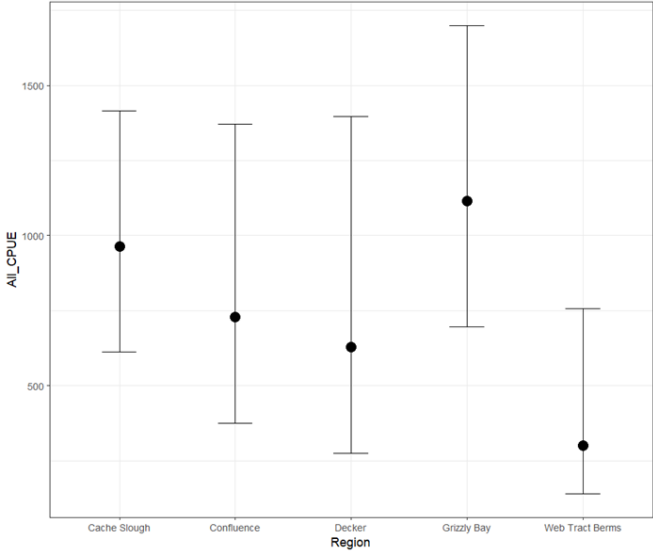
CPUE (all spp): lognormal hurdle

Model 21: All_CPUE ~ Region + (1 | Year)

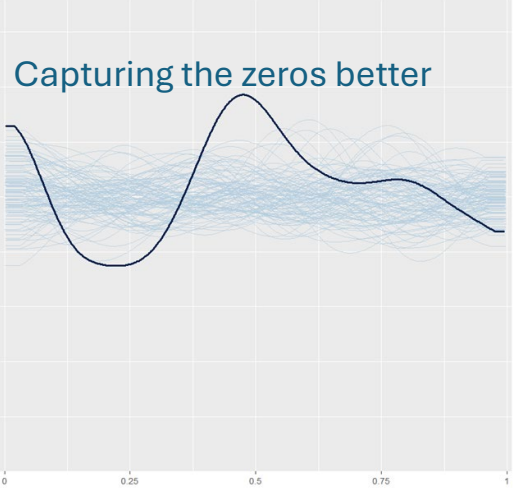
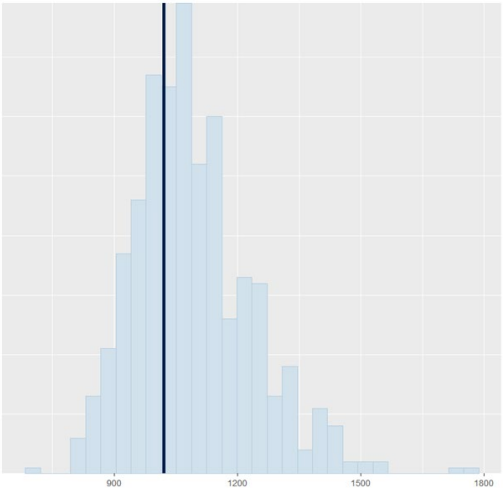
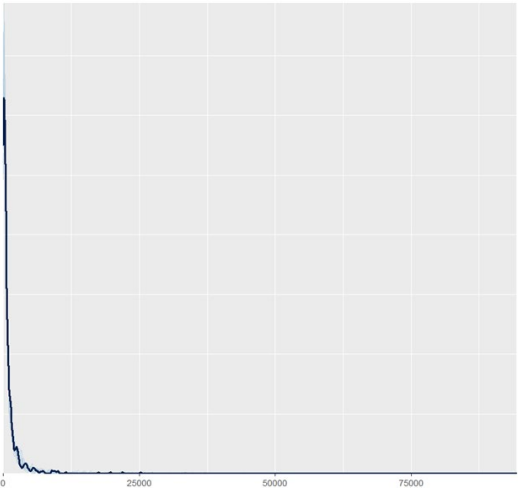
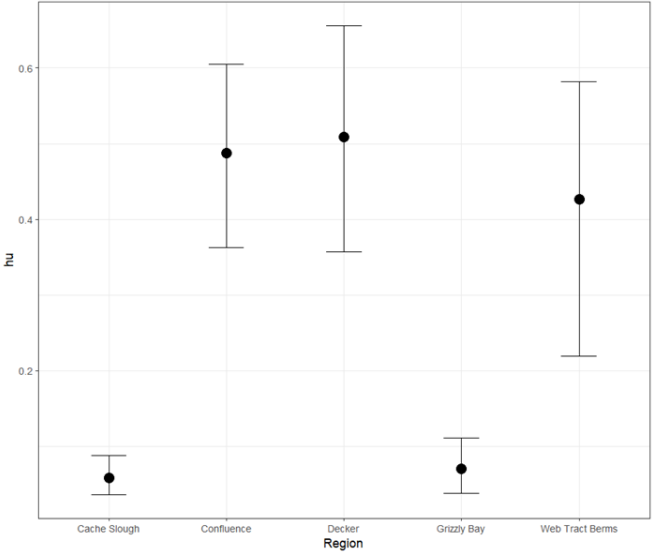
hu ~ Region



Full model (main and hurdle)



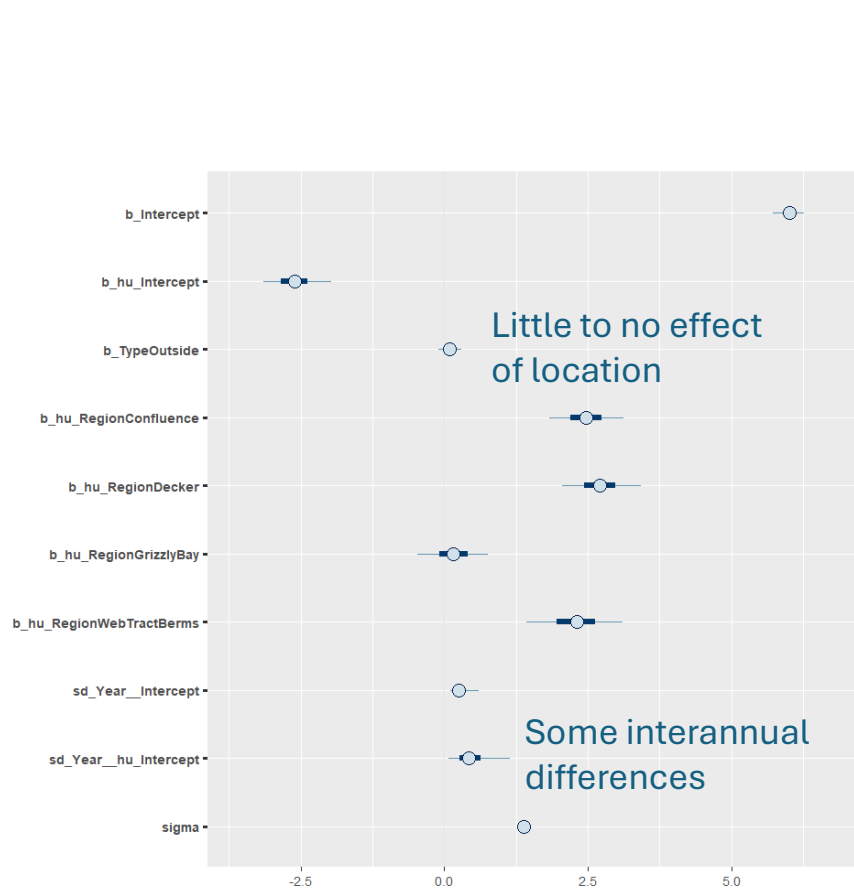
Hurdle only



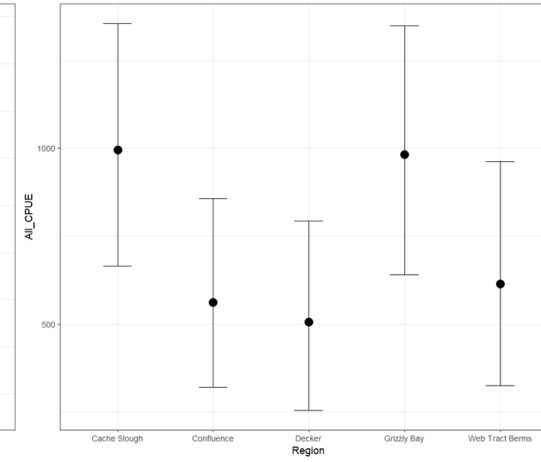
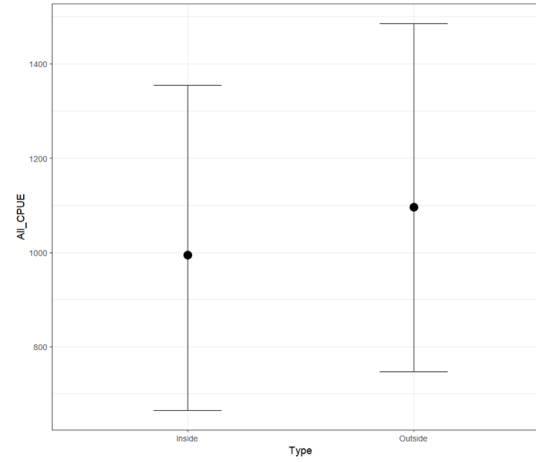
CPUE (all spp): lognormal hurdle

Model 22: All_CPUE ~ Type + (1 | Year)

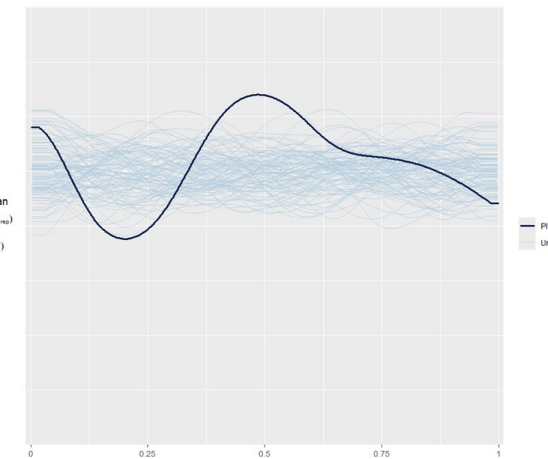
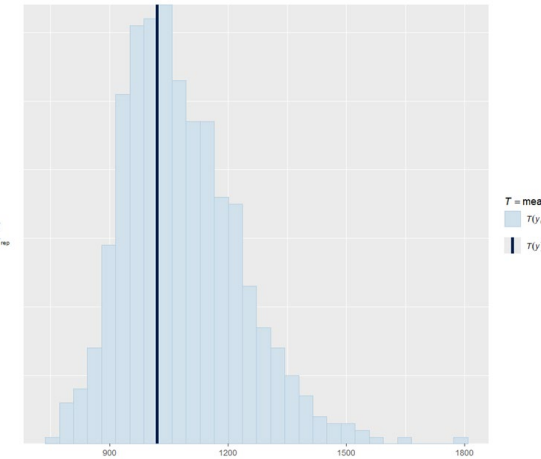
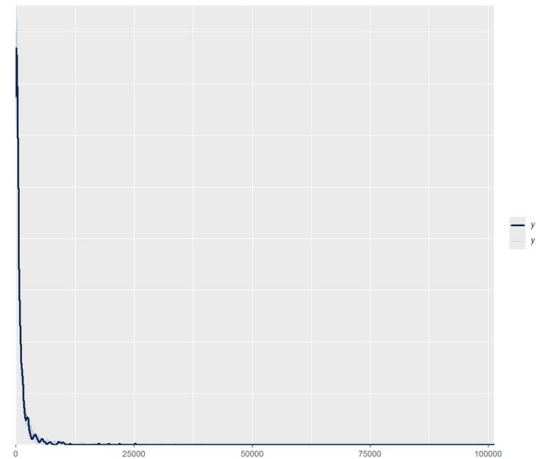
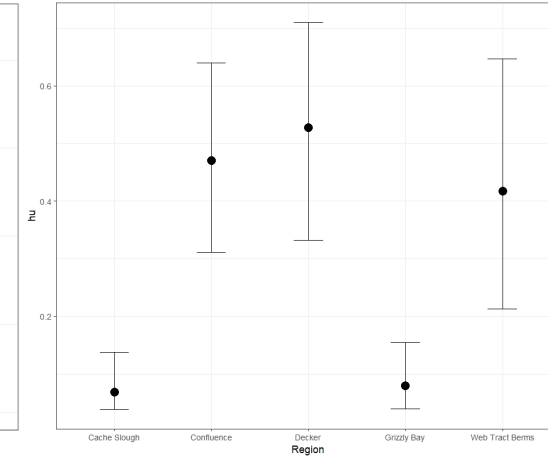
hu ~ Region + (1 | Year)



Full model (main and hurdle)

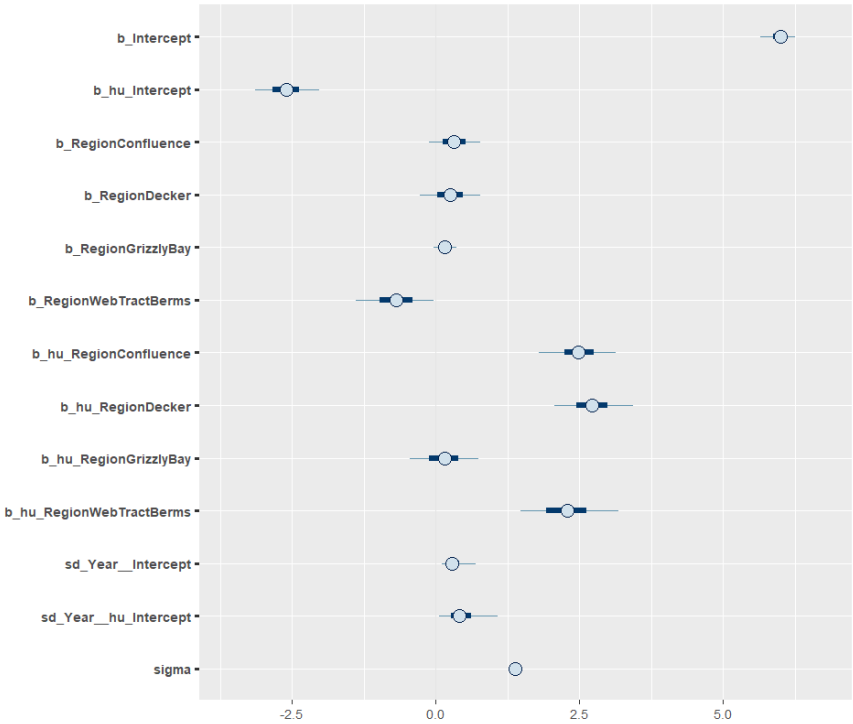


Hurdle only

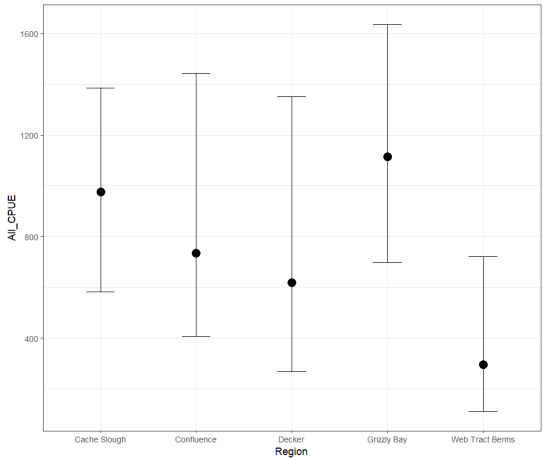


CPUE (all spp): lognormal hurdle

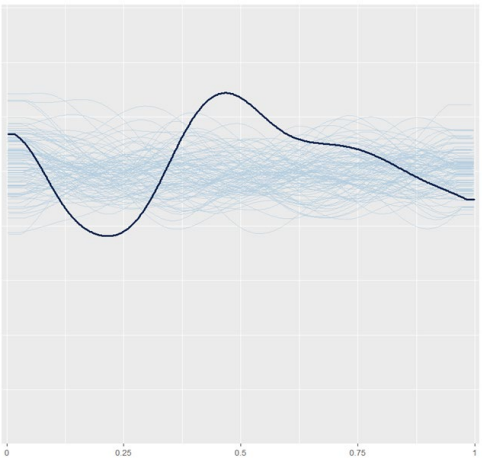
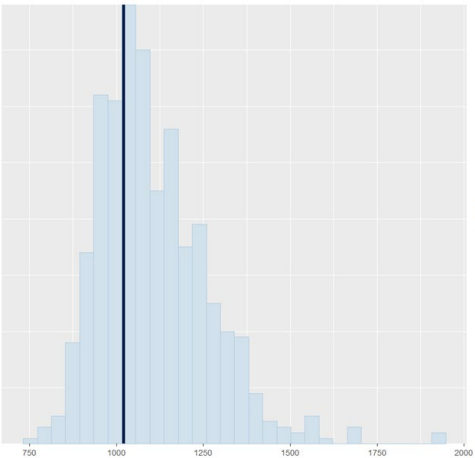
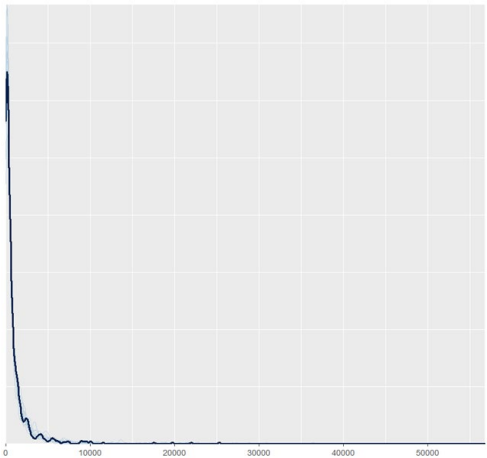
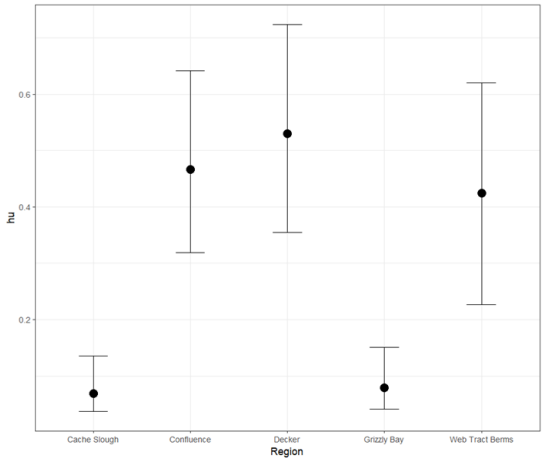
Model 23: All_CPUE ~ Region + (1 | Year)
hu ~ Region + (1 | Year)



Full model (main and hurdle)



Hurdle only



Step 4: Model comparison

CPUE (all spp): all lognormal hurdle models

Models	ELPD_diff	se_diff
23 All_CPUE ~ Region + (1 Year); hu ~ Region + (1 Year)	0.0	0.0
21 All_CPUE ~ (1 Year); hu ~ Region	-0.3	2.3
22 All_CPUE ~ Type + (1 Year), hu ~ Region + (1 Year)	-1.0	2.4
12 All_CPUE ~ Region + (1 Year); hu ~ 1	-57.3	11.4
11 All_CPUE ~ Type + (1 Year) ; hu ~ 1	-58.4	11.6